

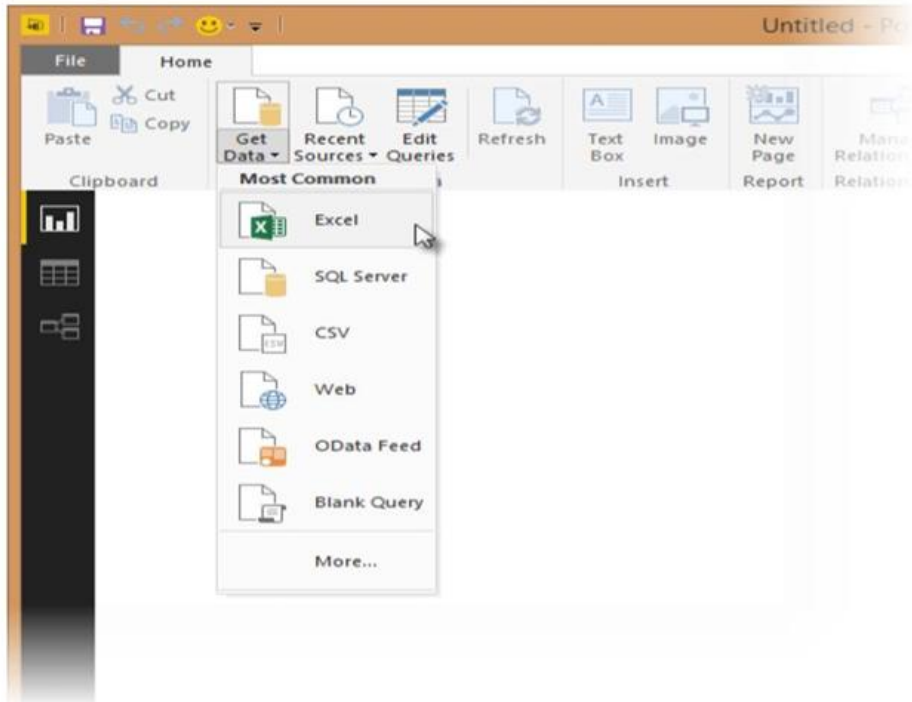
Practical 8- Data Visualization with Power BI

Case Study: Sales Data

Step 1:

Step 1: Connect to an Excel workbook

1. Launch Power BI Desktop.
2. From the Home ribbon, select **Get Data**. Excel is one of the **Most Common** data connections, so you can select it directly from the **Get Data** menu.



3. If you select the Get Data button directly, you can also select **File > Excel** and select **Connect**.
4. In the **Open File** dialog box, select the **Products.xlsx** file.

Step 2. Remove Extra Columns -

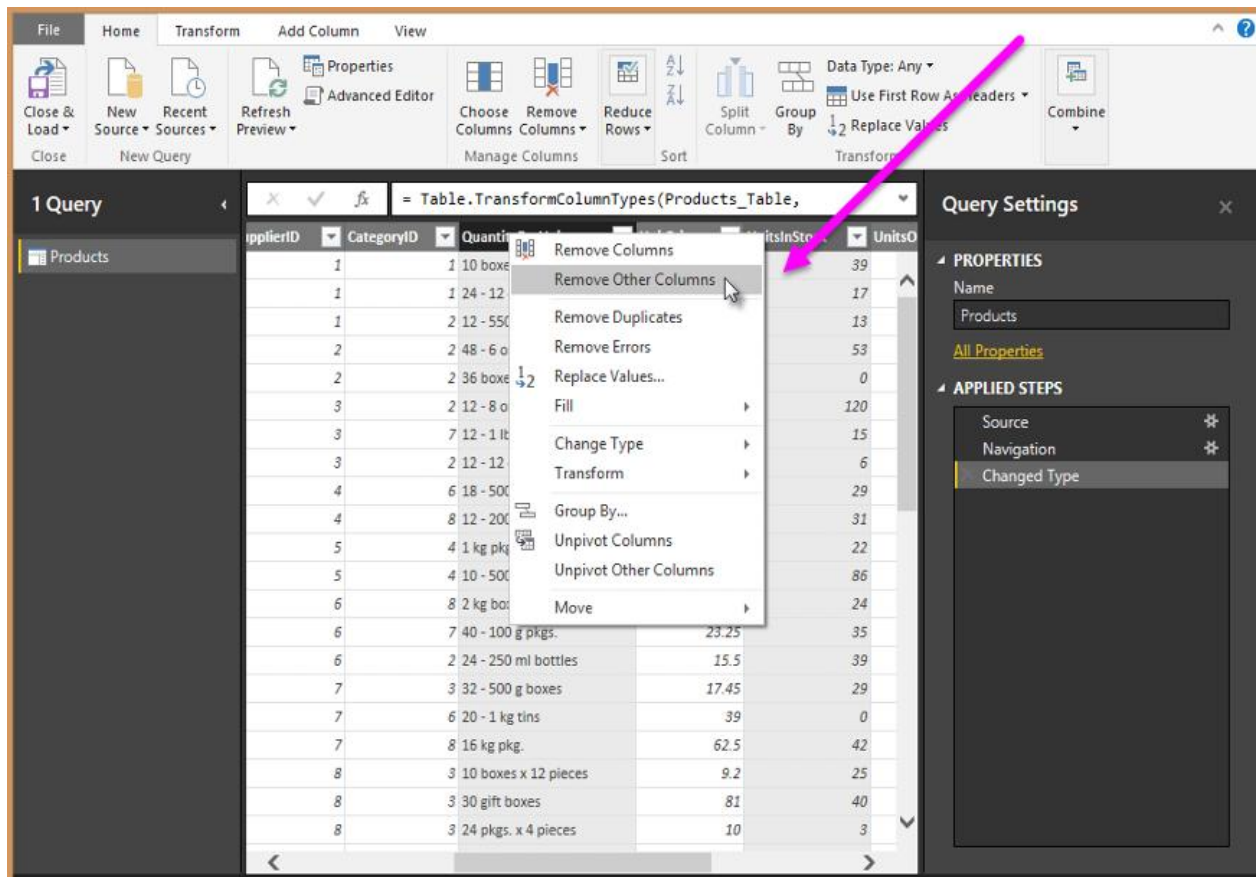
In this step you remove all columns except **ProductID**, **ProductName**, **UnitsInStock**, and **QuantityPerUnit**.

The screenshot shows the Power BI Navigator window with the 'Products' table selected. The table has the following columns: ProductID, ProductName, SupplierID, CategoryID, and QuantityPerUnit. The 'Edit' button is highlighted with a pink arrow.

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit
1	Chai	1	1	10
2	Chang	1	1	24
3	Aniseed Syrup	1	2	12
4	Chef Anton's Cajun Seasoning	2	2	48
5	Chef Anton's Gumbo Mix	2	2	36
6	Grandma's Boysenberry Spread	3	2	12
7	Uncle Bob's Organic Dried Pears	3	7	12
8	Northwoods Cranberry Sauce	3	2	12
9	Mishi Kobe Niku	4	6	18
10	Ikura	4	8	12
11	Queso Cabrales	5	4	1
12	Queso Manchego La Pastora	5	4	10
13	Konbu	6	8	2
14	Tofu	6	7	40
15	Genen Shouyu	6	2	24
16	Pavlova	7	3	32
17	Alice Mutton	7	6	20
18	Carnarvon Tigers	7	8	16
19	Teatime Chocolate Biscuits	8	3	10
20	Sir Rodney's Marmalade	8	3	30
21	Sir Rodney's Scones	8	3	24
22	Gustaf's Knäckebröd	9	5	24

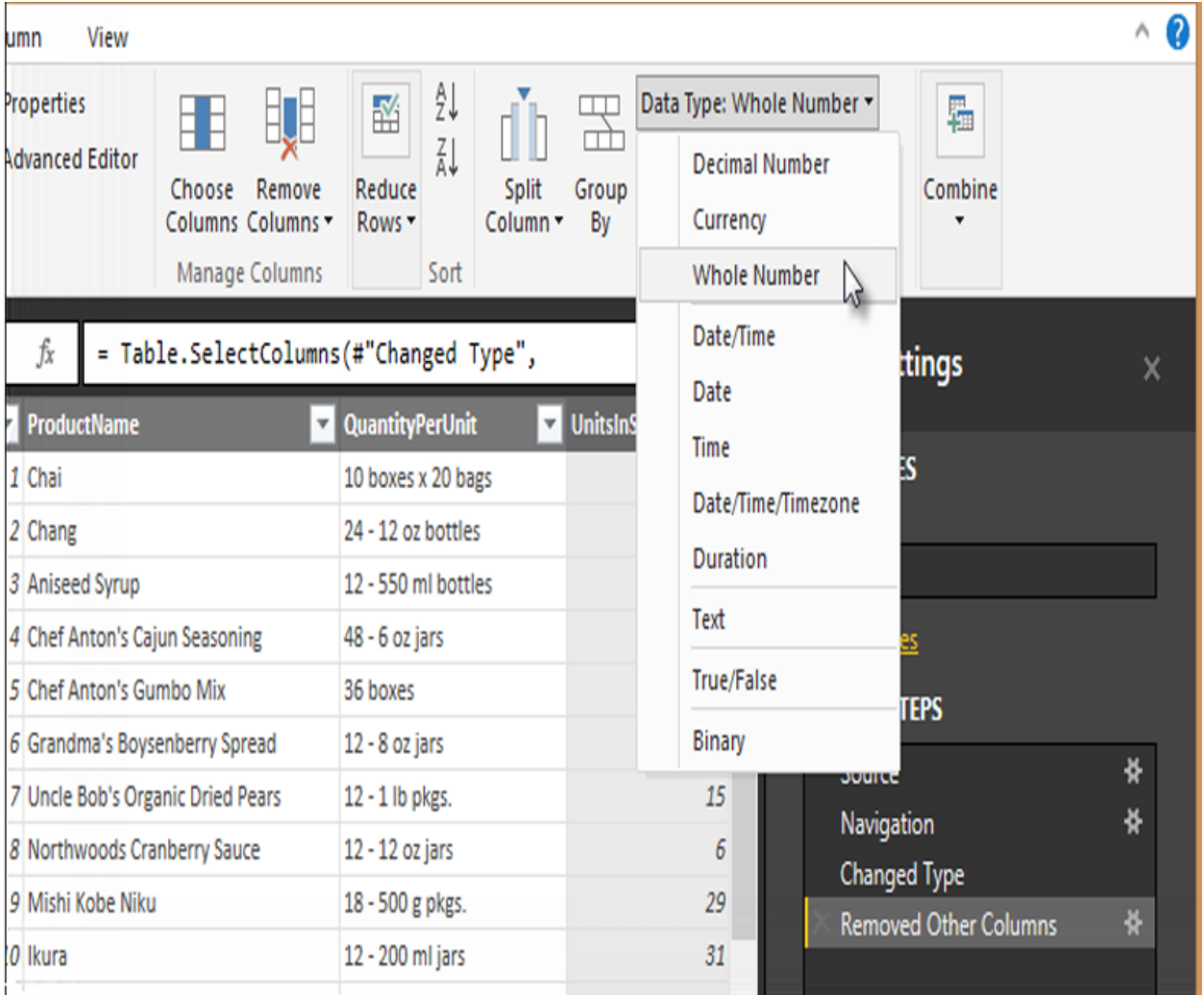
You can also open the Query Editor by selecting Edit Queries from the Home ribbon in Power BI Desktop. The following steps are performed in Query Editor.

- i. In Query Editor, select the ProductID, ProductName, QuantityPerUnit, and UnitsInStock columns (use Ctrl+Click to select more than one column, or Shift+Click to select columns that are beside each other)
- ii. Select Remove Columns - Remove Other Columns from the ribbon, or right-click on a column header and click Remove Other Columns



Step 3: Change the data type of the UnitsInStock column For the Excel workbook, products in stock will always be a whole number, so in this step you confirm the UnitsInStock column's datatype is Whole Number.

- i. Select the UnitsInStock column.
- ii. Select the Data Type drop-down button in the Home ribbon.
- iii. If not already a Whole Number, select Whole Number for data type from the drop down (the Data Type: button also displays the data type for the current selection).



The screenshot shows the Microsoft Excel ribbon with the 'Data Type' dropdown menu open. The menu is currently set to 'Whole Number'. The background shows a table with the following data:

ProductName	QuantityPerUnit	UnitsInStock
1 Chai	10 boxes x 20 bags	
2 Chang	24 - 12 oz bottles	
3 Aniseed Syrup	12 - 550 ml bottles	
4 Chef Anton's Cajun Seasoning	48 - 6 oz jars	
5 Chef Anton's Gumbo Mix	36 boxes	
6 Grandma's Boysenberry Spread	12 - 8 oz jars	
7 Uncle Bob's Organic Dried Pears	12 - 1 lb pkgs.	15
8 Northwoods Cranberry Sauce	12 - 12 oz jars	6
9 Mishi Kobe Niku	18 - 500 g pkgs.	29
10 Ikura	12 - 200 ml jars	31

Task 2: Import order data from an OData feed You import data into Power BI Desktop from the sample Northwind OData feed at the following URL, which you can copy (and then paste) in the steps below:

<http://services.odata.org/V3/Northwind/Northwind.svc/>

Step 1: Connect to an OData feed -

- i. From the Home ribbon tab in Query Editor, select Get Data.
- ii. Browse to the OData Feed data source.
- iii. In the OData Feed dialog box, paste the URL for the Northwind OData feed.
- iv. Select OK.

Navigator

Show All | Show Selected [1]

http://services.odata.org/V3/Northwind/Nort...

- ☐ Alphabetical_list_of_products
- ☐ Categories
- ☐ Category_Sales_for_1997
- ☐ Current_Product_Lists
- ☐ Customer_and_Suppliers_by_Cities
- ☐ CustomerDemographics
- ☐ Customers
- ☐ Employees
- ☐ Invoices
- ☐ Order_Details
- ☐ Order_Details_Extendeds
- ☐ Order_Subtotals
- ☒ Orders
- ☐ Orders_Qries
- ☐ Product_Sales_for_1997
- ☐ Products
- ☐ Products_Above_Average_Prices
- ☐ Products_by_Categories
- ☐ Regions

Orders

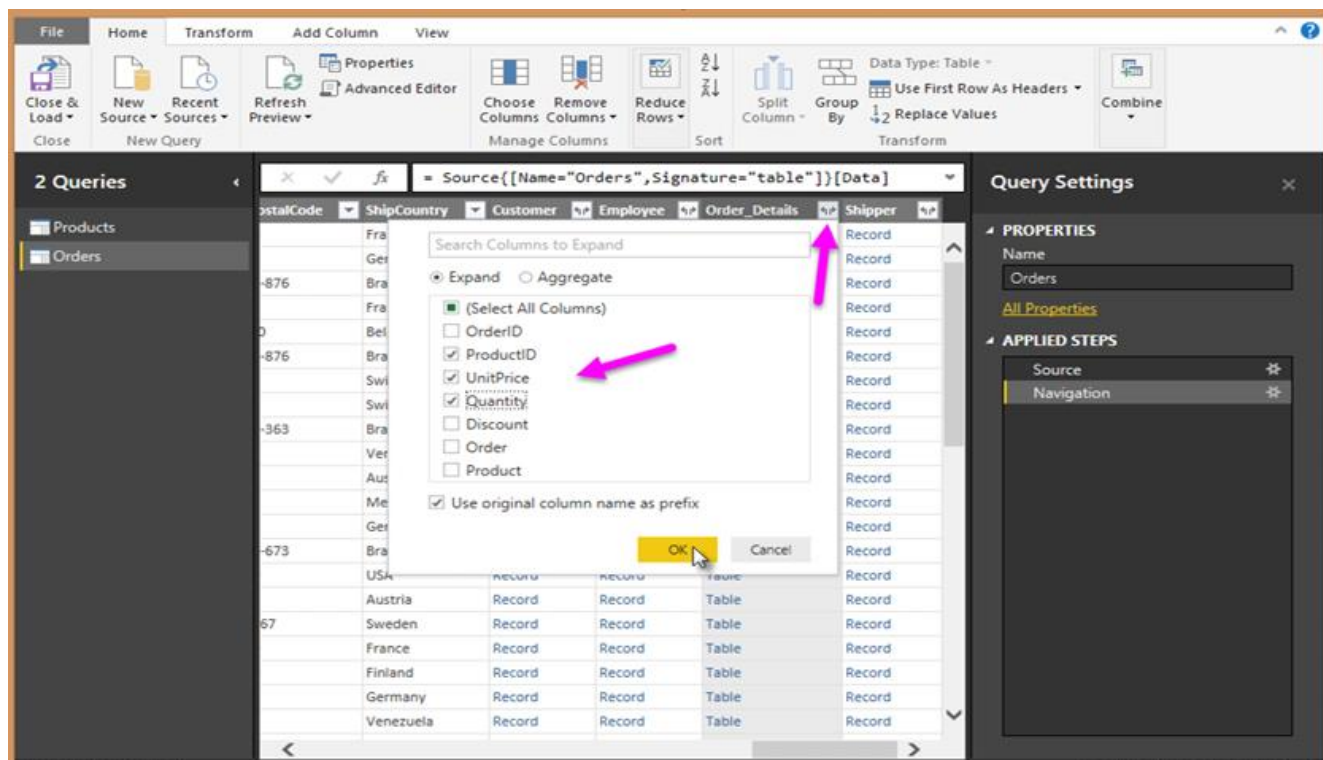
OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate
10248	VINET	5	7/4/1996 12:00:00 AM	8/1/1996
10249	TOMSP	6	7/5/1996 12:00:00 AM	8/16/1996
10250	HANAR	4	7/8/1996 12:00:00 AM	8/5/1996
10251	VICTE	3	7/8/1996 12:00:00 AM	8/5/1996
10252	SUPRD	4	7/9/1996 12:00:00 AM	8/6/1996
10253	HANAR	3	7/10/1996 12:00:00 AM	7/24/1996
10254	CHOPS	5	7/11/1996 12:00:00 AM	8/8/1996
10255	RICSU	9	7/12/1996 12:00:00 AM	8/9/1996
10256	WELLI	3	7/15/1996 12:00:00 AM	8/12/1996
10257	HILAA	4	7/16/1996 12:00:00 AM	8/13/1996
10258	ERNSH	1	7/17/1996 12:00:00 AM	8/14/1996
10259	CENTC	4	7/18/1996 12:00:00 AM	8/15/1996
10260	OTTIK	4	7/19/1996 12:00:00 AM	8/16/1996
10261	QUEDE	4	7/19/1996 12:00:00 AM	8/16/1996
10262	RATTC	8	7/22/1996 12:00:00 AM	8/19/1996
10263	ERNSH	9	7/23/1996 12:00:00 AM	8/20/1996
10264	FOLKO	6	7/24/1996 12:00:00 AM	8/21/1996
10265	BLONP	2	7/25/1996 12:00:00 AM	8/22/1996
10266	WARTH	3	7/26/1996 12:00:00 AM	9/6/1996
10267	FRANK	4	7/29/1996 12:00:00 AM	8/26/1996
10268	GROSR	8	7/30/1996 12:00:00 AM	8/27/1996
10269	WHITC	5	7/31/1996 12:00:00 AM	8/14/1996
10270	WARTH	1	8/1/1996 12:00:00 AM	8/29/1996

OK Cancel

Step 2: Expand the Order_Details table –

Expand the Order_Details table that is related to the Orders table, to combine the ProductID, UnitPrice, and Quantity columns from Order_Details into the Orders table. The Expand operation combines columns from a related table into a subject table. When the query runs, rows from the related table (Order_Details) are combined into rows from the subject table (Orders). After you expand the Order_Details table, three new columns and additional rows are added to the Orders table, one for each row in the nested or related table.

- i. In the Query View, scroll to the Order_Details column.
- ii. In the Order_Details column, select the expand icon ().
- iii. In the Expand drop-down: Select (Select All Columns) to clear all columns.
- iv. Select ProductID, UnitPrice, and Quantity and click OK.



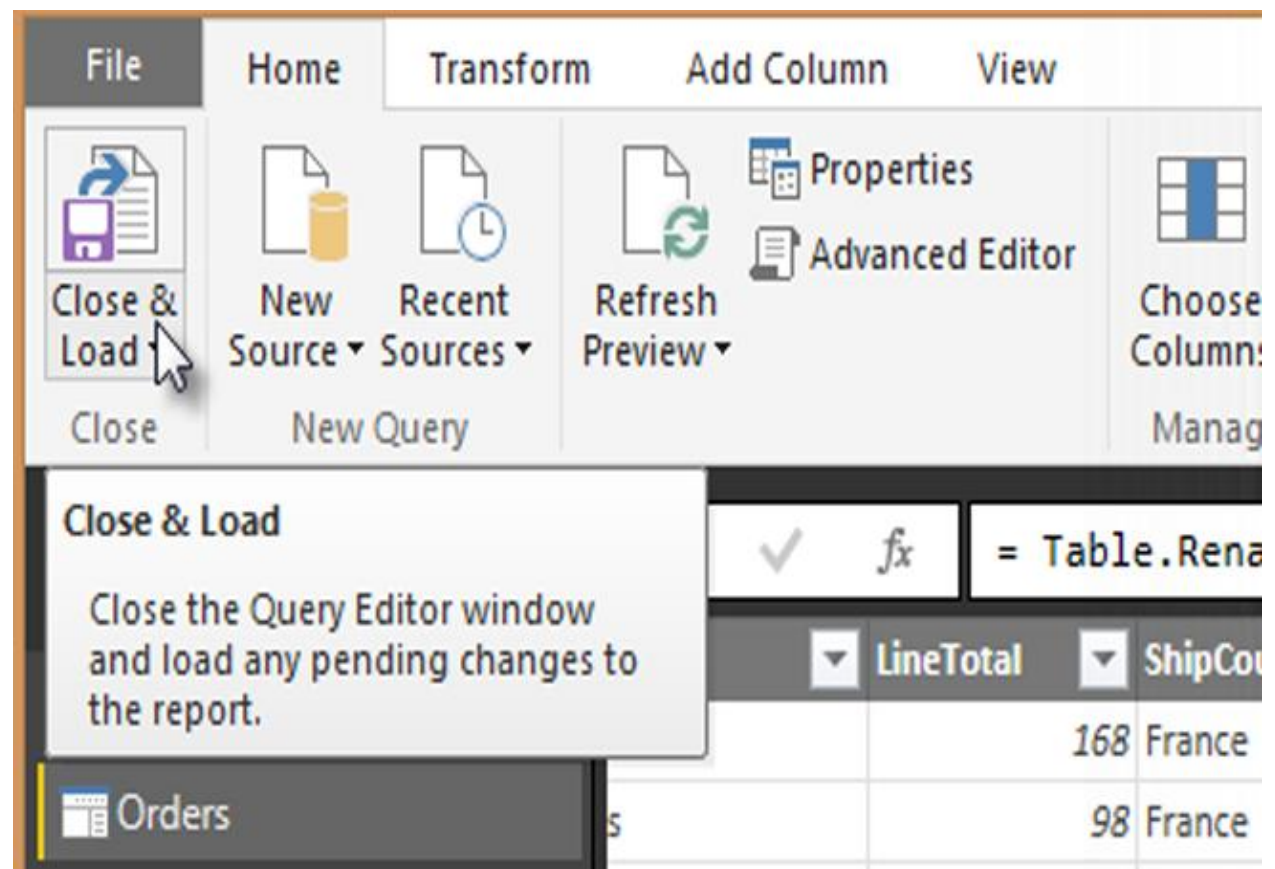
Step 3: Remove other columns to only display columns of interest -

In this step you remove all columns except OrderDate, ShipCity, ShipCountry, Order_Details.ProductID, Order_Details.UnitPrice, and Order_Details.Quantity columns. In the previous task, you used Remove Other Columns.

For this task, you remove selected columns.

- i. In the Query View, select all columns by completing point 'iii' of Step 2.
- ii. Click the first column (OrderID).
- iii. Shift+Click the last column (Shipper).
- iv. Now that all columns are selected, use Ctrl+Click to unselect the following columns: OrderDate, ShipCity, ShipCountry, Order_Details.ProductID, Order_Details.UnitPrice, and Order_Details.Quantity.
- v. Now that only the columns we want to remove are selected, right-click on any selected column header and click Remove Columns.

Now as this Steps are done click on Close and Load to Load the Changes Properly



Step 4: Calculate the line total for each Order_Details row - Power BI Desktop lets you create calculations based on the columns you are importing, so you can enrich the data that you connect to.

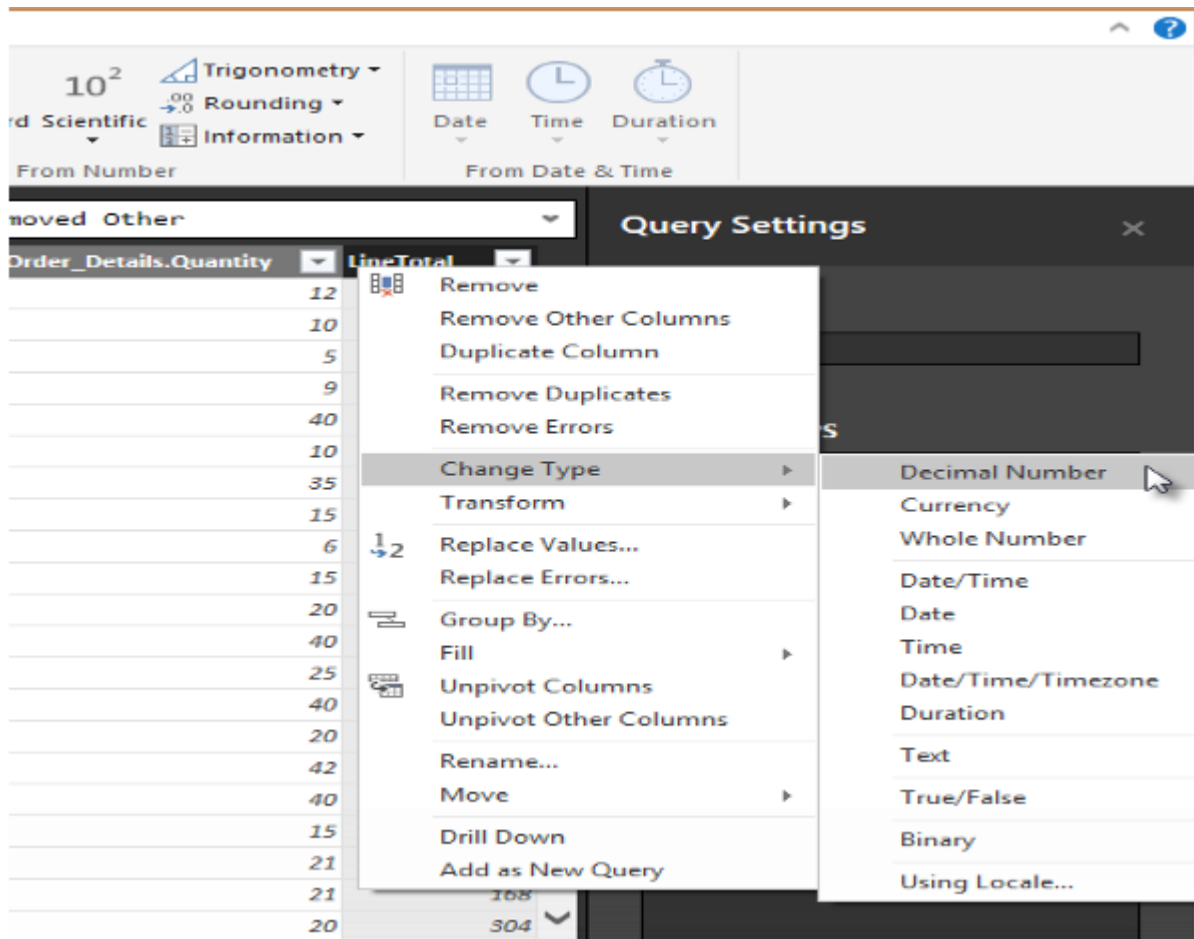
In this step, you create a Custom Column to calculate the line total for each Order_Details row. Calculate the line total for each Order_Details row:

- i. In the Add Column ribbon tab, click Add Custom Column.**
- ii. In the Add Custom Column dialog box, in the Custom Column Formula textbox, enter `[Order_Details.UnitPrice] * [Order_Details.Quantity]`.**
- iii. In the New column name textbox, enter LineTotal.**

The screenshot shows the 'Add Custom Column' dialog box. The 'New column name' field is filled with 'LineTotal'. The 'Custom column formula' field contains the DAX formula `= [Order_Details.UnitPrice] * [Order_Details.Quantity]`. On the right, the 'Available columns' list shows several fields, with 'Order_Details.Quantity' highlighted. Below the list is an '<< Insert' button. At the bottom left, a green checkmark indicates 'No syntax errors have been detected.' At the bottom right, there are 'OK' and 'Cancel' buttons. A red arrow points from the 'OK' button to the text 'Click OK.' on the right side of the page.

Click OK.

- Step 5: Set the datatype of the LineTotal field –**
- i. Right click the LineTotal column.
 - ii. Select Change Type and choose Decimal Number.



Step 6: Rename and reorder columns in query

- In Query Editor, drag the LineTotal column to the left, after ShipCountry.
- Remove the Order_Details. prefix from the Order_Details.ProductID, Order_Details.UnitPrice and Order_Details.Quantity columns, by double-clicking on each column header, and then deleting that text from the column name.

The screenshot shows the Power BI Query Editor interface. On the left, the '2 Queries' pane lists 'Products' and 'Orders'. The main area displays a table with the following columns: ShipCity, LineTotal, Order_Details.ProductID, and Order_Details.UnitPrice. The 'LineTotal' column is being dragged to the left of the 'ShipCity' column. The 'Query Settings' pane on the right shows the 'PROPERTIES' section with the 'Name' set to 'Orders'. The 'APPLIED STEPS' section lists the following steps: Source, Navigation, Expanded Order_Details, Removed Other Columns, and Added Custom.

ShipCity	LineTotal	Order_Details.ProductID	Order_Details.UnitPrice
AM Reims	France	11	42
AM Reims	France	72	14
AM Münster	Germany	51	41
AM Münster	Germany	51	65
AM Rio de Janeiro	Brazil	22	57
AM Rio de Janeiro	Brazil	65	65
AM Lyon	France	20	33
AM Lyon	France	60	31
AM Lyon	France	39	49
AM Charleroi	Belgium	24	55
AM Charleroi	Belgium	74	2
AM Charleroi	Belgium		
AM Rio de Janeiro	Brazil		
AM Rio de Janeiro	Brazil		
AM Rio de Janeiro	Brazil		
AM Bern	Switzerland		
AM Bern	Switzerland		
AM Bern	Switzerland		
AM Genève	Switzerland		

7 COLUMNS, 999+ ROWS

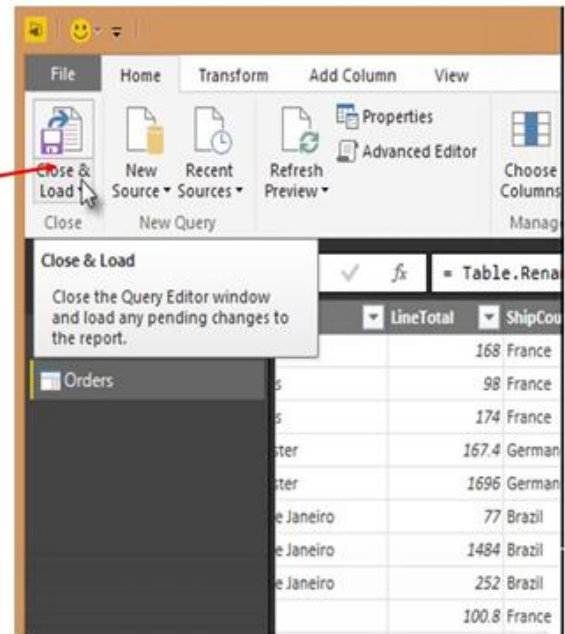
PREVIEW DOWNLOADED AT 9:52 AM

Task 3: Combine the Products and Total Sales queries

Step 1:

Step 1: Confirm the relationship between Products and Total Sales

1. First, we need to load the model that we created in Query Editor into Power BI Desktop. From the **Home** ribbon of Query Editor, select **Close & Load**.

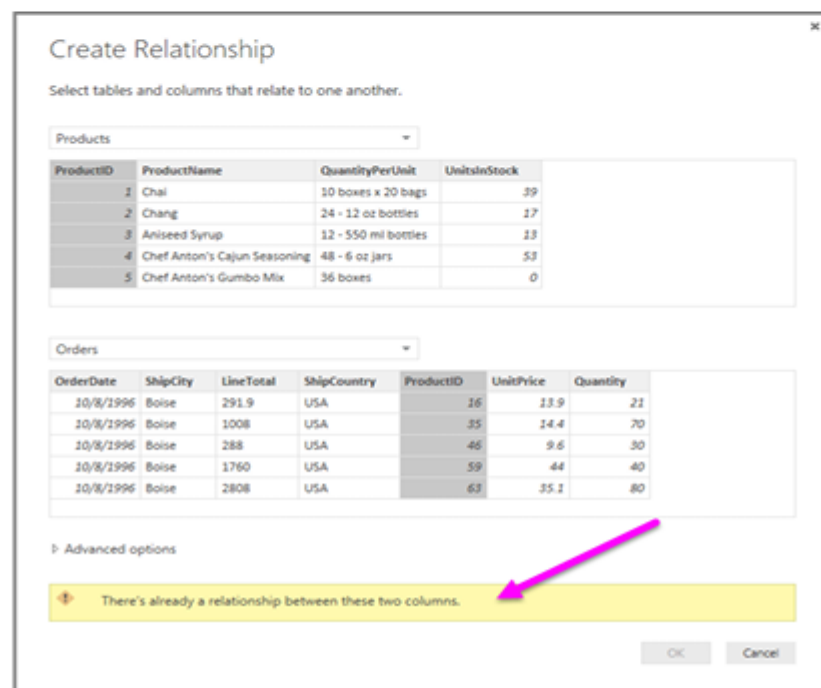
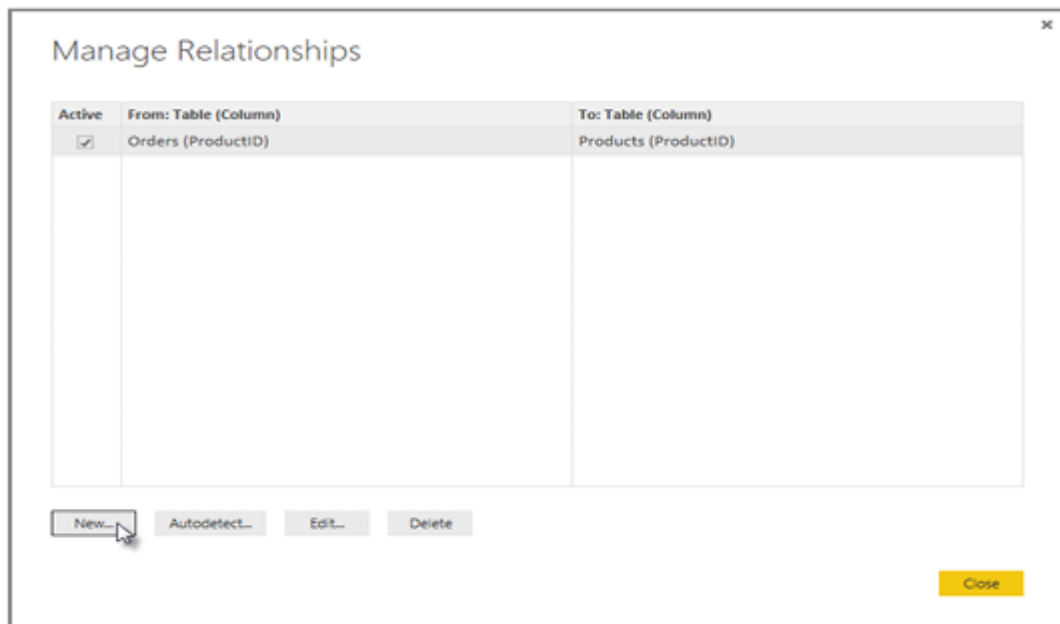


Step 2. Power BI Desktop loads the data from the two queries

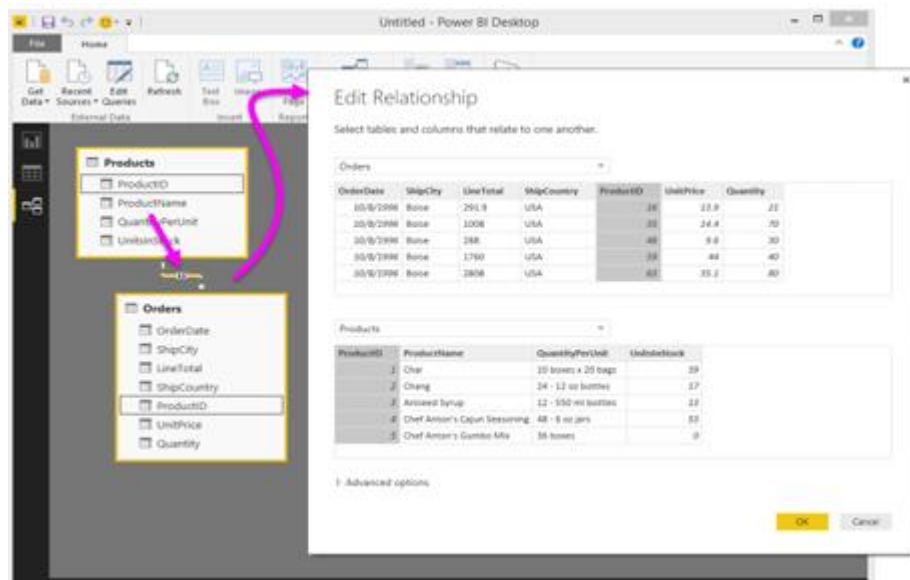
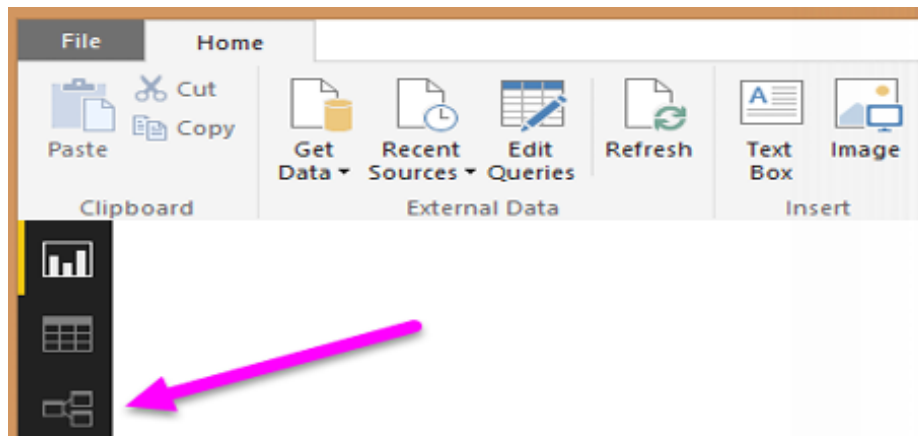
Step 3. Once the data is loaded, select the Manage Relationships button Home ribbon

Step 4. Select the New... button

Step 5. When we attempt to create the relationship, we see that one already exists!
As shown in the Create Relationship dialog (by the shaded columns), the ProductsID fields in each query already have an established relationship.

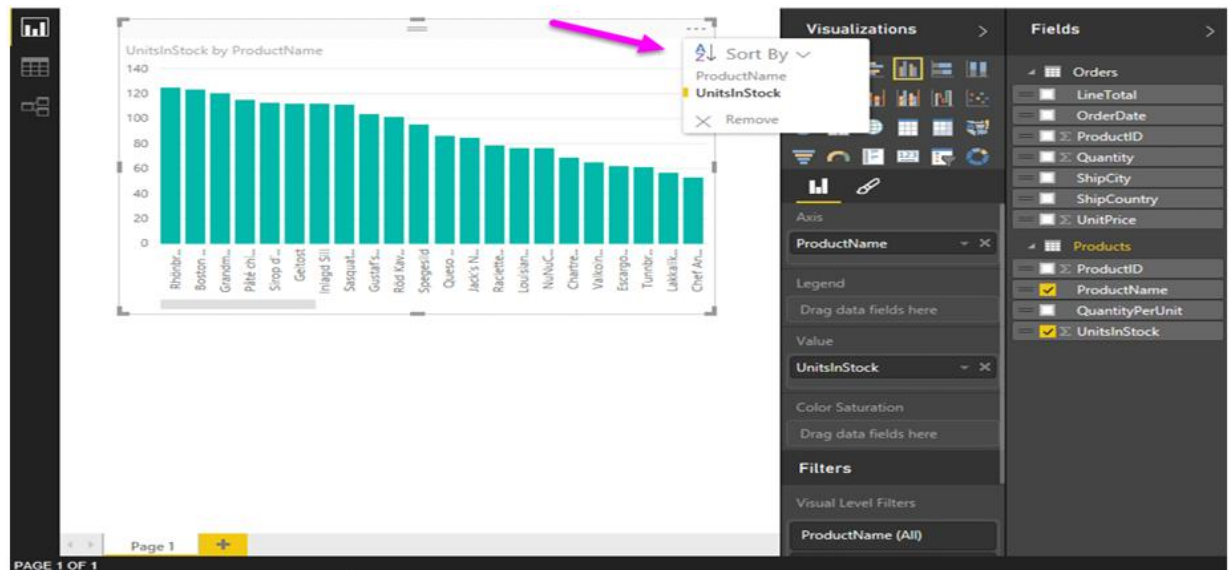


Step 6. Select Cancel, and then select Relationship view in Power BI Desktop.

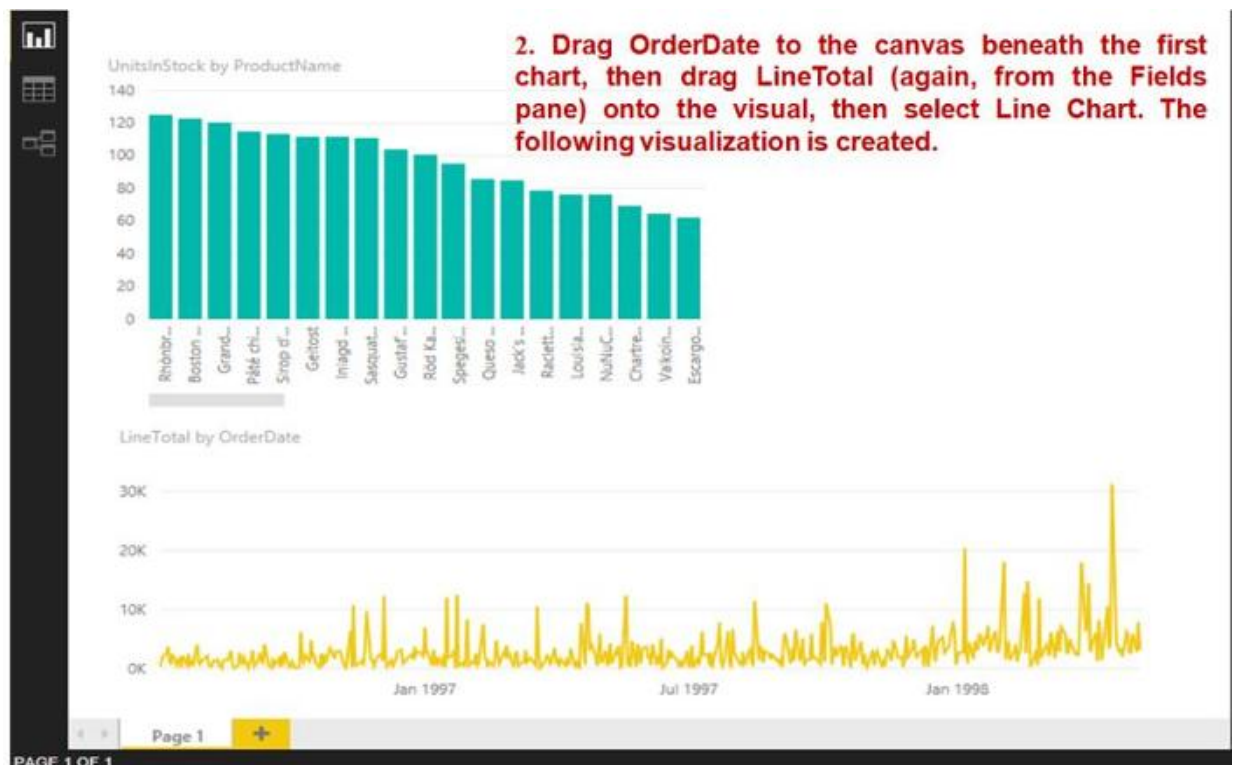


Task 4: Build visuals using your data

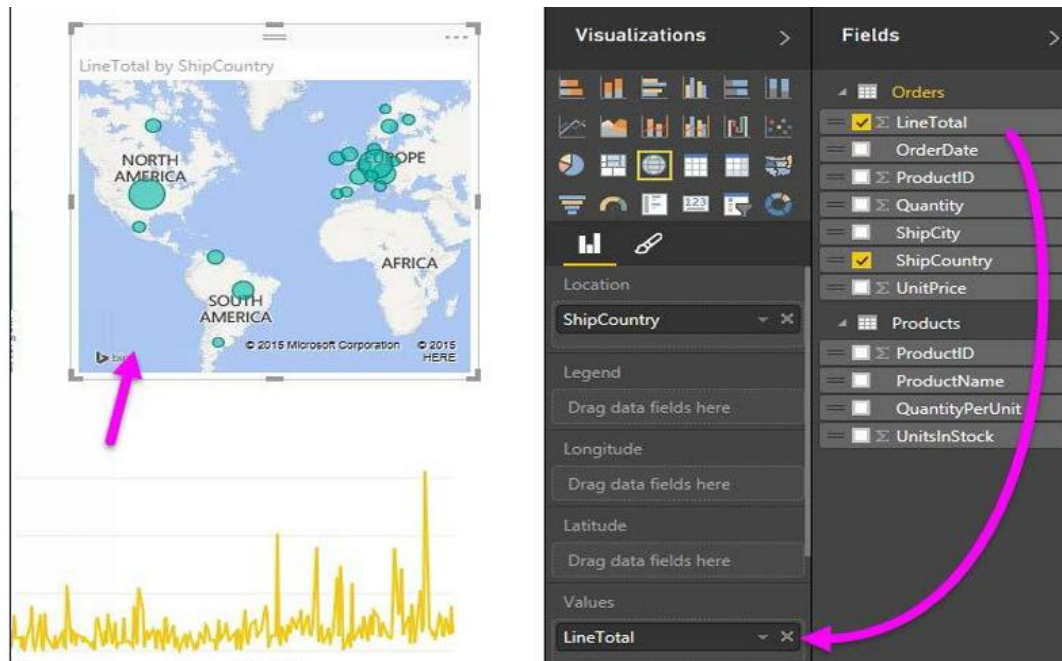
Step 1: Create charts showing Units in Stock by Product and Total Sales by Year



Step 2:



Step 3: Next, drag ShipCountry to a space on the canvas in the top right. Because you selected a geographic field, a map was created automatically. Now drag LineTotal to the Values field; the circles on the map for each country are now relative in size to the LineTotal for orders shipped to that country.



Step 4: Interact with your visuals to analyze further

